Not All Label Errors Are Equal: Efficient Task-Aware Auditing with Confidence Sequences

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Abstract

We study efficient, adaptive auditing of categorical labels in a fixed dataset, motivated by biodiversity-credit programs in which expert verification of species records is costly and downstream scores (i.e., payment) are nonlinear functions of the corrected labels. We represent label corrections with a submitted-to-verified confusion matrix. For each row (signifying a class of submitted labels), we construct a confidence sequence under randomized sampling without replacement, and aggregate these row-wise confidence sequences into a confidence sequence for the full matrix. Projecting this confidence sequence through the downstream scoring functions then yields a confidence sequence for the score. To extend this result into a practical auditing framework, we then develop a task-aware adaptive policy that prioritizes which row (e.g., species class) to inspect next based on their remaining uncertainty and their influence on the downstream score. We show that adaptive row selection and data-dependent stopping preserve simultaneous $1-\alpha$ coverage of the downstream score, provided records are sampled randomly within each row. Together, these results provide a general framework for anytime-valid, decision-aware auditing of labels in finite datasets.

1 Problem and contribution

Large labeled datasets support consequential evaluation and scientific decisions, yet expert verification of every label is often infeasible. Existing confidence sequences for sampling without replacement certify simple finite-population quantities [Waudby-Smith and Ramdas, 2020, Ryu and Wornell, 2024]; active evaluation directs labels toward model-risk estimation [Sawade et al., 2012, Kossen et al., 2021, 2022, Mittal et al., 2024], and recent work combines active acquisition with anytime-valid risk control [Xu et al., 2024]. Our setting couples these problems: submitted labels may be wrong, errors are categorical and directional, and the downstream target may be nonlinear or nonsmooth.

Motivation: Auditing for biodiversity credits Biodiversity credits are purchased by companies and investors to fund nature conservation and restoration worldwide, compensating individual landowners for measurable improvements in ecosystem health [Wunder et al., 2025]. The market is estimated to grow from US$8B annually to $180B by 2050 [World Economic Forum, 2023], but credits are burdened by high overhead costs and concerns about trust and verification [Aide, 2024]. Each project may generate tens or hundreds of thousands of sensor records documenting wildlife presence and abundance, making full expert review impractical.

Automated species classifiers reduce labeling cost but can exhibit substantial error: one widely used classifier had an 18.2% error rate on an out-of-sample dataset [Tabak et al., 2019]. Human labels are also imperfect; in a study of snow-leopard identification, nonexperts had a 14.6% misclassification rate, inflating estimated abundance by roughly 35% [Johansson et al., 2020].

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Because credit payments depend on these biodiversity metrics, label errors can have direct financial consequences, creating a need for efficient expert auditing. An aggregate mislabel rate (e.g., "13% of labels are incorrect") is insufficient for this task because the *direction* of an error determines its downstream consequence. Incorrectly labeling cattle as lion, for example, may falsely introduce a rare species and substantially inflate estimated richness and taxonomic diversity (and therefore payment), whereas confusing two common, closely related species may have little effect. An effective audit should therefore account not only for where errors are likely, but also for how potential label corrections affect the downstream scoring function.

The same structure arises in myriad settings. In model evaluation, for example, noisy benchmark labels impact downstream quantities such as F_1 score, balanced accuracy, or model rankings, each of which may depend nonlinearly on the corrected labels. In epidemiology, label corrections may change prevalence estimates, while in content moderation they may alter policy-compliance rates.

Our approach Here, we study adaptive label verification for a fixed finite dataset when the downstream scoring function is a nonlinear and non-smooth function of the corrected labels. Our approach constructs anytime-valid confidence intervals for this scoring function while also deciding which records to verify next. Anytime validity allows the auditor to stop as soon as a claim is certified, rejected, or estimated with sufficient precision, without fixing the audit size in advance.

We make three methodological contributions. First, we construct an anytime-valid confidence sequence for the submitted-to-verified confusion matrix of a fixed finite dataset. Second, we propagate this structured uncertainty through nonlinear downstream scoring functions, obtaining anytime-valid confidence sequences for the downstream scores together with Lipschitz-based upper bounds on their widths. Third, we use row-level uncertainty and downstream sensitivity to guide adaptive auditing while preserving validity under data-dependent row selection and stopping.

On a replay of 33,403 records from a biodiversity credit project in central Kenya, the priority-aware auditing rule introduced in Section D achieved a mean Pillar 2 (defined later) absolute-error tolerance of 0.0426 using approximately 2,520 expert audits, compared with 5,000 audits under simple random sampling of the next row. In this experiment, the priority-aware method audited 7.5% rather than 15.0% of the dataset, saving approximately 2,480 audits.

Setup. Record $i \in [N]$ has submitted label $y_i \in \mathcal{L} = \{1, \ldots, L\}$ and (initially unknown) verified label v_i. Define

$$C^\star_{rs} = \#\{i : y_i = r, v_i = s\}, \qquad N_r = \sum_s C^\star_{rs}, \tag{1}$$

where N_r is known. Any candidate C induces corrected abundance $m_s(C) = \sum_r C_{rs}$, composition $p_s(C) = m_s(C)/N$, and a deterministic score $g(C)$. In the application, g includes species richness P_1, within-group Hill–Shannon diversity $P_2 = \sum_G \exp\{-\sum_{s\in G} q_{G,s} \log q_{G,s}\}$, taxonomic diversity P_3, and a multimetric payment that aggregates P_1, P_2, and P_3. Full definitions and sensitivity bounds are in Appendix A.

2 Anytime-valid matrix certification

Within row r, audits reveal a uniformly randomized order without replacement. After u audits let $A_{rs,u}$ be the number verified as s. For a candidate total row $q \in \mathbb{Z}_+^L$, put a strictly positive Dirichlet–multinomial *working* prior $\pi_{r,0}$ on q and let $\pi_{r,u}$ be its posterior after $A_{r,u}$. The prior is only a device for constructing the e-process; no generative model is assumed. Define

$$R_{r,u}(q) = \frac{\pi_{r,0}(q)}{\pi_{r,u}(q)}, \qquad \widetilde{\mathcal{C}}_{r,u}(\alpha_r) = \bigcap_{j\le u} \left\{ q : \sum_s q_s = N_r,\ q_s \ge A_{rs,j},\ R_{r,j}(q) < \alpha_r^{-1} \right\}. \tag{2}$$

Theorem 1 (Anytime-valid correction and score sets)**.** *Choose $\alpha_r > 0$ with $\sum_r \alpha_r \le \alpha$. At each global time, select a row predictably from the complete audit history and reveal the next record in that row's independently randomized order. Then*

$$\mathbb{P}\Big(C^\star \in \mathcal{C}_t^C := \{C : C_{r\cdot} \in \widetilde{\mathcal{C}}_{r,u_r(t)}(\alpha_r)\ \forall r\}\ \forall t\Big) \ge 1 - \alpha. \tag{3}$$

Consequently, for every deterministic score g, $I_t(g) = [\min_{C\in\mathcal{C}_t^C} g(C), \max_{C\in\mathcal{C}_t^C} g(C)]$ covers $g(C^\star)$ simultaneously for all t, including every data-dependent stopping time.

The proof combines the row PPR nonnegative supermartingale, Ville's inequality, and a union bound; projection is deterministic. The complete closed form and proofs are in Appendices B–C. The theorem permits adaptive row choice but not targeted record choice within a row. Previously verified targeted records may be conditioned on and removed before randomizing the residual population.

3 Priority-aware sampling

Validity leaves open where to audit. Let $\mathcal{P}_t = \{p(C) : C \in \mathcal{C}_t^C\}$, $w_r = N_r/N$, $S_{r,t}$ be plausible verified destinations from row r, and $\delta_{r,u}$ the ℓ_1 diameter of its row set. For score g_j and a predeclared full-width tolerance τ_j, define

$$\Lambda_{jr,t} = \frac{w_r}{2\tau_j} \sup_{p \in \mathcal{P}_t} \left[\max_{s \in S_{r,t}} \partial_s g_j(p) - \min_{s \in S_{r,t}} \partial_s g_j(p) \right]. \tag{4}$$

Theorem 2 (Directional width and allocation)**.** *On a convex differentiability region,* $\mathrm{width}\{I_t(g_j)\}/\tau_j \le \sum_r \Lambda_{jr,t}\delta_{r,u_r(t)}$. *For frozen nonnegative coefficients* λ_r *and decreasing convex planning radii* $\bar{\delta}_r(u)$*, repeatedly choosing* $r \in \arg\max_k \lambda_k\{\bar{\delta}_k(u_k) - \bar{\delta}_k(u_k+1)\}$ *globally minimizes* $\sum_r \lambda_r \bar{\delta}_r(u_r)$ *at every feasible integer budget. If* $\bar{\delta}_r(u) \simeq \kappa_r u^{-1/2}$*, then* $u_r^* \propto (\lambda_r\kappa_r)^{2/3}$.

The theorem optimizes a tractable surrogate, not the random exact nonlinear projection width. Proofs and extensions to heterogeneous costs, approximate priorities, and richness are in Appendices D and F.

Implementable support-aware PAS. A plug-in rule can assign zero priority to unseen but consequential transitions. We instead estimate (4) by posterior draws from a positive Dirichlet–multinomial working model, so every unseen transition retains positive posterior probability. We begin with $m_r = \min\{N_r, \max(5, \lceil\sqrt{N_r}\rceil)\}$ randomized audits per row, update at geometric epochs, and freeze coefficients within each epoch. An SRS floor ε chooses row r with probability

$$\varepsilon \frac{N_r - u_r(t)}{N - t} + (1-\varepsilon)\mathbb{1}\{r = r_t^{\mathrm{PAS}}\}. \tag{5}$$

Because row selection is predictable and the next within-row record remains randomized, any such priority estimator, epoch schedule, or ε preserves Theorem 1. Auditors stop only when a projected interval certifies/rejects a threshold or has width at most error tolerance τ_j.

4 Experiments

The dataset from a biodiversity credit project in central-Kenya contains 33,403 expert-verified records with 46 submitted species, 32 verified species, 50 species in their union, and 6,002 corrections (17.97%). Rows range from 2 to 12,075 records; 20 have fewer than 20 records, while five transitions contain 64.1% of corrections. We replay 100 paired within-row random orders through 5,000 audits. Table 1 separates point-estimation and offline allocation diagnostics; only the separately evaluated PPR projection is a confidence sequence.

Table 1: Evaluation on Kenya dataset at 5,000 audits (mean of 100 paired paths; lower is better). Surrogate and TV are offline diagnostics.

Policy	Surrogate	Pillar 2 MAE	Total Variation
Stratified Random Sampling (SRS)	28.60	0.0426	0.516
Plug-in Priority-Aware Sampling (PAS)	29.41	**0.0203**	0.215
Support-presence PAS	**18.77**	0.0279	**0.140**
Submitted-catalog PAS	22.12	0.0328	0.328
PAS, 75% SRS floor	23.15	0.0347	0.403
PAS, 90% SRS floor	25.05	0.0353	0.458

Support-presence PAS reduces the common truth-evaluated surrogate by 34.4%, Pillar 2 MAE by 34.5%, and allocation total variation distance by 72.9% versus SRS. Using SRS's 5,000-audit mean

MAE, 0.0426 effective-species units, as a retrospective matched-accuracy tolerance, support-presence PAS reaches it after about 2,520 audits: 7.5% rather than 15.0% of the population, **saving 2,480 audits (49.6%)**. Because true MAE is unavailable mid-audit, this is not a live certification rule.

Coverage and certified-width tests. Across 12 finite populations (36,000 complete audit paths), nominal 95% PPR constructions have worst full-path noncoverage 3.17% (joint product) and 2.47% (rowwise Bonferroni); repeatedly viewed fixed-time Wald intervals fail on 71.1% of paths. In an eight-row, 5,700-record heterogeneous-influence simulation with a 20-per-row pilot, PAS reaches 35% of the common post-pilot exact designed-score width in 533 audits versus median 996 for random rows (46.5% fewer); the local envelope is 3.7% wider than the exact interval at stopping. Gains fall to 2% under homogeneous influence. In an exactly enumerated $N = 24$ categorical benchmark, the true Pillar 2 is 2.937; at half census the mean exact PPR width is 0.867 and the envelope 1.870, both collapsing to zero at census. Designs, uncertainty bands, negative controls, and figures appear in Appendix E.

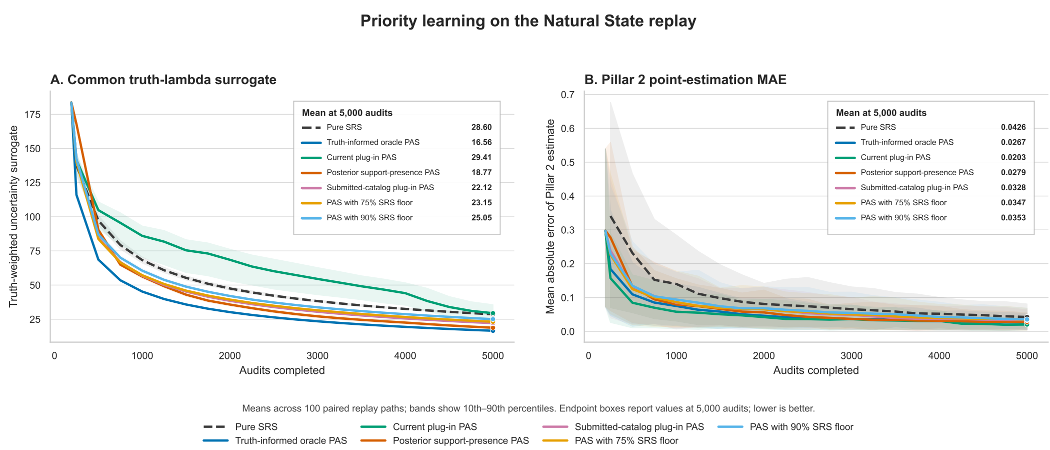

Figure 1: Evaluation on real-world dataset over 100 paired paths. Panel A is a common truth-evaluated, non-certified planning surrogate; Panel B is Pillar-2 point-estimation MAE. Bands are 10th–90th percentiles. Full interpretation appears in Appendix E.3.

5 Conclusion

We develop an anytime-valid method for auditing categorical datasets when label errors affect biodiversity-credit decisions. The method builds a confidence sequence for the full confusion matrix—which submitted labels truly correspond to which verified labels—and propagates this uncertainty to richness, diversity, and taxonomic-diversity scores. Our task-aware policy audits the submitted-label row expected to reduce uncertainty in the downstream score most. It accounts for the row's size, its remaining uncertainty, and the possible consequences of its label corrections. When these influence weights are fixed, the greedy rule is optimal for the proposed planning objective.

The results show when priority-aware auditing is useful. In the 33,403-record evaluation on data from a real biodiversity project, posterior support-presence allocation reached the same point-estimation accuracy as simple random sampling with 5,000 audits after about 2,520 audits, a retrospective reduction of 49.6%. Simulations similarly show substantial gains when correction directions have very different effects on the target score. When rows have similar downstream importance, however, priority-aware sampling provides little advantage. Exact projection of the high-dimensional confusion-matrix confidence set also remains computationally demanding.

Overall, task-aware auditing is most useful when different possible label corrections have meaningfully different consequences for the final decision. Future work should develop sharper and faster confidence intervals for the downstream scores, reducing the conservatism and computational cost of projection.

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A Expanded problem formulation and biodiversity scores

This supplement preserves the modeling, theoretical, and empirical detail omitted from the four-page paper. All labels and complete-dataset transitions are fixed and non-random; probability refers only to the auditor's randomized inspection order.

B Anytime-Valid Confusion-Matrix Certification

We construct an anytime-valid confidence sequence for the full correction matrix. The submitted labels induce a natural stratification: each submitted-label row is a fixed finite population, and the auditor samples uniformly without replacement within whichever row is selected. The auditor may choose rows adaptively using all information revealed so far.

Let $\mathcal{L} = \{1, \ldots, L\}$ denote the label set and let

$$C^\star_{rs} := \#\{i : y_i = r,\ v_i = s\}, \qquad r, s \in \mathcal{L}, \tag{6}$$

be the unknown confusion matrix of the complete dataset. Its row sums $N_r := \sum_s C^\star_{rs}$ are known from the submitted labels. Every candidate matrix C deterministically induces corrected species abundances and hence every downstream pillar or payment score. We first construct a categorical confidence sequence for one row, combine the row sets into a joint matrix confidence sequence, and then project that set through the frozen scoring program.

B.1 A categorical confidence sequence for one row

Fix a submitted-label row r. Its unknown verified-label count vector is

$$Q^\star_r := C^\star_{r\cdot} = (C^\star_{r1}, \ldots, C^\star_{rL}), \qquad \sum_{s=1}^{L} C^\star_{rs} = N_r. \tag{7}$$

Suppose that u records have been sampled uniformly without replacement from this row. Let $V_{r,u} \in \mathcal{L}$ be the verified label in the u-th row-local audit and define

$$A_{rs,u} := \sum_{j=1}^{u} \mathbb{1}\{V_{r,j} = s\}, \qquad A_{r,u} := (A_{r1,u}, \ldots, A_{rL,u}). \tag{8}$$

With $\mathcal{F}_{r,u} := \sigma(V_{r,1}, \ldots, V_{r,u})$, sampling without replacement implies

$$\mathbb{P}(V_{r,u+1} = s \mid \mathcal{F}_{r,u}) = \frac{C^\star_{rs} - A_{rs,u}}{N_r - u}. \tag{9}$$

Choose a strictly positive working vector $\beta_r = (\beta_{r1}, \ldots, \beta_{rL})$, put $B_r := \sum_s \beta_{rs}$, and place the Dirichlet–multinomial working prior

$$Q_r \sim \mathrm{DirMult}(N_r, \beta_r) \tag{10}$$

on the complete row. This prior is used only to construct the e-process; it is not assumed to generate the finite population. Let $\pi_{r,0}(q)$ be its prior mass and $\pi_{r,u}(q)$ its posterior mass after observing $A_{r,u}$. Define the prior–posterior ratio

$$R_{r,u}(q) := \frac{\pi_{r,0}(q)}{\pi_{r,u}(q)}. \tag{11}$$

Lemma 1 (Row PPR supermartingale; Waudby-Smith and Ramdas, 2020)**.** *Under the sampling-without-replacement law of the true row, the process $\{R_{r,u}(Q^\star_r)\}_{u=0}^{N_r}$ is a nonnegative supermartingale with $R_{r,0}(Q^\star_r) = 1$.*

For $\alpha_r \in (0, 1)$, define the raw PPR set

$$\mathcal{C}^{\mathrm{raw}}_{r,u}(\alpha_r) := \left\{ q \in \mathbb{Z}^L_{\geq 0} : q_s \geq A_{rs,u}\ \forall s, \quad \sum_{s=1}^{L} q_s = N_r, \quad R_{r,u}(q) < \frac{1}{\alpha_r} \right\} \tag{12}$$

and its running intersection

$$\widetilde{\mathcal{C}}_{r,u}(\alpha_r) := \bigcap_{j=0}^{u} \mathcal{C}^{\mathrm{raw}}_{r,j}(\alpha_r). \tag{13}$$

Theorem 3 (One-row categorical confidence sequence)**.** *For every $\alpha_r \in (0,1)$,*

$$\mathbb{P}\big\{Q_r^\star \in \mathcal{C}^{\mathrm{raw}}_{r,u}(\alpha_r) \text{ for every } u = 0,\dots,N_r\big\} \geq 1-\alpha_r. \tag{14}$$

Consequently, $\{\widetilde{\mathcal{C}}_{r,u}(\alpha_r)\}_{u=0}^{N_r}$ is a nested anytime-valid confidence sequence. At a full row audit, $\widetilde{\mathcal{C}}_{r,N_r}(\alpha_r) \subseteq \{Q_r^\star\}$, and it equals $\{Q_r^\star\}$ on the simultaneous coverage event in (14).

For computation, the PPR has the closed form

$$R_{r,u}(q) = \frac{N_r!}{(N_r-u)!}\frac{\Gamma(B_r)}{\Gamma(B_r+u)}\prod_{s=1}^{L}\frac{(q_s - A_{rs,u})!}{q_s!}\frac{\Gamma(\beta_{rs}+A_{rs,u})}{\Gamma(\beta_{rs})}, \tag{15}$$

for every feasible q with $q_s \geq A_{rs,u}$ and $\sum_s q_s = N_r$. We evaluate $\log R_{r,u}(q)$ with log-gamma functions. Appendix C proves Lemma 1, derives (15), and proves Theorem 3.

B.2 From row sets to a confusion-matrix confidence sequence

Let $u_r(t)$ be the number of randomized audits completed in row r by global audit time t. Choose row error budgets $\alpha_r > 0$ satisfying $\sum_r \alpha_r \leq \alpha$, and define

$$\mathcal{C}_t^C := \left\{C : C_{r\cdot} \in \widetilde{\mathcal{C}}_{r,u_r(t)}(\alpha_r) \text{ for every row } r\right\}. \tag{16}$$

Theorem 4 (Confusion-matrix confidence sequence)**.** *Pre-randomize the remaining records independently within each submitted-label row. At each global audit time, the auditor may select the next row predictably using the complete audit history and then reveal the next unseen record in that row's randomized order. Under this design,*

$$\mathbb{P}\{C^\star \in \mathcal{C}_t^C \text{ for every global time } t\} \geq 1-\alpha. \tag{17}$$

In particular, $C^\star \in \mathcal{C}_\tau^C$ with probability at least $1-\alpha$ for every data-dependent stopping time τ.

The theorem permits adaptive row choice but not targeted choice of an individual record within a row. Previously verified records selected by an arbitrary mechanism may be conditioned upon: remove their known transitions from the residual finite population, apply the randomized procedure to the remaining records, and add the known transition counts to every candidate matrix. The resulting set still covers the complete matrix. The proof is in Appendix C.

B.3 Projection to downstream scores

For a deterministic, frozen scoring program g, define

$$I_t(g) := \left[\min_{C\in\mathcal{C}_t^C} g(C),\ \max_{C\in\mathcal{C}_t^C} g(C)\right]. \tag{18}$$

Theorem 5 (Projection confidence sequence)**.** *For every deterministic score g, including Pillars P_1, P_2, P_3 and a fixed implementation of the official payment software,*

$$\mathbb{P}\{g(C^\star) \in I_t(g) \text{ for every } t\} \geq 1-\alpha. \tag{19}$$

Thus $I_\tau(g)$ remains valid at every data-dependent stopping time τ.

Projection is valid for nonlinear and non-smooth scores, although projection of a joint confidence set may be conservative [Kaido et al., 2019]. The following deterministic bridge gives a convenient width certificate for smooth composition-dependent scores.

For a candidate matrix C, let

$$\theta_r(C) := \frac{C_{r\cdot}}{N_r}, \qquad w_r := \frac{N_r}{N}, \qquad p(C) := \left(\frac{1}{N}\sum_r C_{r1},\dots,\frac{1}{N}\sum_r C_{rL}\right). \tag{20}$$

Then

$$p(C) = \sum_r w_r\theta_r(C). \tag{21}$$

Define the row diameter after u audits by

$$\delta_{r,u} := \sup_{q,q' \in \widetilde{\mathcal{C}}_{r,u}(\alpha_r)} \left\| \frac{q}{N_r} - \frac{q'}{N_r} \right\|_1, \tag{22}$$

whenever the displayed row set is nonempty. All subsequent width statements are interpreted on the simultaneous coverage event, on which every displayed set is nonempty. Equation (21) implies

$$\mathrm{diam}_1\{p(C) : C \in \mathcal{C}_t^C\} \le \sum_r w_r \delta_{r,u_r(t)}. \tag{23}$$

Proposition 1 (Lipschitz bridge)**.** *Suppose that h satisfies*

$$|h(p) - h(p')| \le L_h \|p - p'\|_1 \tag{24}$$

throughout $\{p(C) : C \in \mathcal{C}_t^C\}$. Then its exact projected interval obeys

$$\mathrm{width}\{I_t(h)\} \le L_h\, \mathrm{diam}_1\{p(C) : C \in \mathcal{C}_t^C\} \le L_h \sum_r w_r \delta_{r,u_r(t)}. \tag{25}$$

B.4 Specialization to the biodiversity pillars

Pillar 1 and the presence-dependent part of Pillar 3 are non-smooth and should be handled by exact projection or verified-presence witnesses. Pillar 2 and the abundance-weighted component of Pillar 3 are smooth away from support and denominator boundaries, where the following local bounds apply.

Pillar 1: species richness. Let $\mathcal{I}_s$ be a prespecified candidate pool of n_s records that could contain verified species s. A single verified occurrence certifies presence in the complete dataset. Once such an occurrence is found, further audits have zero marginal value for the richness-only objective of establishing species s's presence, although they may remain valuable for other pillars.

Proposition 2 (Verified-presence and first-hit guarantee)**.** *Let $L_{1,t}$ be the number of distinct species observed among verified records by time t. Then $L_{1,t} \le P_1(C^\star)$ pathwise. If $\mathcal{I}_s$ contains $m_s \ge 1$ true occurrences of species s and is sampled uniformly without replacement, then after u audits*

$$\mathbb{P}(T_s \le u) = 1 - \frac{\binom{n_s - m_s}{u}}{\binom{n_s}{u}}, \qquad \mathbb{E}[T_s] = \frac{n_s + 1}{m_s + 1}, \tag{26}$$

where T_s is the audit index of the first verified occurrence. Non-detection does not certify absence unless all records that could contain species s have been excluded or audited.

A submitted-label row may serve as $\mathcal{I}_s$ only for detecting species s within that row. Because misclassification can create incoming transitions from other rows, certifying global absence requires exact projection over all rows or an independently justified candidate pool covering every possible occurrence.

Pillar 2: Hill–Shannon diversity. For target group G, let $m_G(p) := \sum_{s \in G} p_s$ and let $q_G(p) := p_G / m_G(p)$ be its normalized composition. On an active support of size d_G, define

$$H(q) := -\sum_{s \in G} q_s \log q_s, \qquad D_1(q) := \exp\{H(q)\}. \tag{27}$$

Theorem 6 (Pillar 2 Lipschitz constant)**.** *Let $\mathcal{P}$ be a region of corrected compositions such that, for group G, the active support is fixed with size d_G, $m_G(p) \ge \mu_G > 0$, and every active coordinate of $q_G(p)$ is at least $\eta_G > 0$ for every $p \in \mathcal{P}$. Then*

$$|D_1(q_G(p)) - D_1(q_G(p'))| \le L_{2,G} \|p - p'\|_1, \tag{28}$$

$$L_{2,G} := \frac{2 d_G}{\mu_G} \max\{1, |1 + \log \eta_G|\}. \tag{29}$$

If $P_2(p) = \sum_G D_1(q_G(p))$, one valid global constant is

$$L_2 := \sum_G L_{2,G}. \tag{30}$$

At a support boundary, Shannon diversity remains continuous but is not globally Lipschitz. The audit should then use exact projection or a continuity modulus rather than (29).

Pillar 3: abundance-weighted taxonomic diversity. Within a normalized target group, let q be the species composition, let $0 \le \omega_{ij} \le \Omega$ be the symmetric taxonomic distance with $\omega_{ii} = 0$, and define

$$A(q) := \sum_{i<j} \omega_{ij} q_i q_j, \qquad B(q) := \sum_{i<j} q_i q_j, \qquad \Delta(q) := \frac{A(q)}{B(q)}. \tag{31}$$

Theorem 7 (Smooth Pillar 3 Lipschitz constant)**.** *On any region of normalized compositions satisfying $B(q) \ge \beta > 0$,*

$$|\Delta(p) - \Delta(q)| \le L_3 \|p - q\|_1, \qquad L_3 := \frac{2\Omega}{\beta}. \tag{32}$$

Presence-based and richness-weighted components of Pillar 3 remain non-smooth and must be projected exactly or handled through the same verified-presence logic as Pillar 1.

Smooth multimetric uncertainty. For

$$M_{\text{sm}}(C) := \lambda_2 P_2(C) + \lambda_3 P_3^{\text{sm}}(C), \tag{33}$$

the preceding constants yield the following audit-time certificate.

Corollary 1 (Smooth multimetric width)**.** *On a plausible composition region satisfying the assumptions of Theorems 6 and 7,*

$$\text{width}\{I_t(M_{\text{sm}})\} \le (|\lambda_2| L_2 + |\lambda_3| L_3) \sum_r w_r \delta_{r,u_r(t)}. \tag{34}$$

For a nonlinear payment transformation, one may bound its Jacobian over the current plausible pillar region and compose that bound with (34). Exact projection of the frozen payment program remains the final certificate whenever it is computationally feasible.

C Proofs for Section B

C.1 Row PPR process and one-row confidence sequence

Proof of Lemma 1. Suppress the row subscript r. Conditional on $\mathcal{F}_u$, let

$$p_u^\star(s) := \mathbb{P}_{Q^\star}(V_{u+1} = s \mid \mathcal{F}_u) = \frac{Q_s^\star - A_{s,u}}{N - u} \tag{35}$$

be the true without-replacement predictive probability, and let

$$\widehat{p}_u(s) := \sum_q \pi_u(q) \frac{q_s - A_{s,u}}{N - u} \tag{36}$$

be the working posterior predictive probability. Bayes' rule gives, for every label that can occur under the true row,

$$\frac{R_{u+1}(Q^\star)}{R_u(Q^\star)} = \frac{\widehat{p}_u(V_{u+1})}{p_u^\star(V_{u+1})}. \tag{37}$$

Therefore

$$\mathbb{E}_{Q^\star}\left[\frac{R_{u+1}(Q^\star)}{R_u(Q^\star)} \,\middle|\, \mathcal{F}_u\right] = \sum_{s:p_u^\star(s)>0} p_u^\star(s) \frac{\widehat{p}_u(s)}{p_u^\star(s)} \tag{38}$$

$$= \sum_{s:p_u^\star(s)>0} \widehat{p}_u(s) \le 1. \tag{39}$$

The inequality can be strict because the working predictive distribution may assign mass to a label whose remaining true count is zero. Multiplying by the nonnegative, $\mathcal{F}_u$-measurable quantity $R_u(Q^\star)$ yields

$$\mathbb{E}_{Q^\star}[R_{u+1}(Q^\star) \mid \mathcal{F}_u] \le R_u(Q^\star). \tag{40}$$

Finally, $\pi_0/\pi_0 = 1$, so $R_0(Q^\star) = 1$. Hence the process is a nonnegative supermartingale. $\square$

Derivation of Equation (15). Again suppress the row subscript. Under the Dirichlet–multinomial working prior, the prior mass of a complete row q is

$$\pi_0(q) = \frac{N!}{\prod_s q_s!} \frac{\Gamma(B)}{\Gamma(B+N)} \prod_{s=1}^{L} \frac{\Gamma(\beta_s + q_s)}{\Gamma(\beta_s)}. \tag{41}$$

Given q, the unordered count vector A_u from a uniform sample of size u has multivariate-hypergeometric likelihood

$$\ell_u(A_u; q) = \frac{\prod_s \binom{q_s}{A_{s,u}}}{\binom{N}{u}}. \tag{42}$$

The prior predictive mass of A_u is Dirichlet–multinomial:

$$m_u(A_u) = \frac{u!}{\prod_s A_{s,u}!} \frac{\Gamma(B)}{\Gamma(B+u)} \prod_s \frac{\Gamma(\beta_s + A_{s,u})}{\Gamma(\beta_s)}. \tag{43}$$

Bayes' rule gives $\pi_u(q) = \pi_0(q)\ell_u(A_u; q)/m_u(A_u)$, and hence

$$R_u(q) = \frac{m_u(A_u)}{\ell_u(A_u; q)}. \tag{44}$$

Substituting (42) and (43), cancelling $u!$ and $A_{s,u}!$, and using $\binom{q_s}{A_{s,u}} = q_s!/[A_{s,u}!(q_s - A_{s,u})!]$ gives

$$R_u(q) = \frac{N!}{(N-u)!} \frac{\Gamma(B)}{\Gamma(B+u)} \prod_s \frac{(q_s - A_{s,u})!}{q_s!} \frac{\Gamma(\beta_s + A_{s,u})}{\Gamma(\beta_s)}, \tag{45}$$

which is (15). □

Proof of Theorem 3. By Lemma 1 and Ville's inequality,

$$\mathbb{P}\left\{ \sup_{0 \le u \le N_r} R_{r,u}(Q_r^\star) \ge \frac{1}{\alpha_r} \right\} \le \alpha_r. \tag{46}$$

On the complementary event, the true row satisfies $R_{r,u}(Q_r^\star) < 1/\alpha_r$ for every u. It also satisfies $C_{rs}^\star \ge A_{rs,u}$ for every s, u and $\sum_s C_{rs}^\star = N_r$. Hence $Q_r^\star \in \mathcal{C}_{r,u}^{\mathrm{raw}}(\alpha_r)$ for every u, proving (14). Membership in every raw set implies membership in every running intersection. Nestedness follows directly from $\widetilde{\mathcal{C}}_{r,u+1} = \widetilde{\mathcal{C}}_{r,u} \cap \mathcal{C}_{r,u+1}^{\mathrm{raw}}$.

At $u = N_r$, all row records have been observed, so $A_{r,N_r} = Q_r^\star$. Any feasible candidate q must satisfy $q_s \ge Q_{rs}^\star$ for every s while both vectors sum to N_r; therefore $q = Q_r^\star$. Thus the final running intersection is either empty or the singleton $\{Q_r^\star\}$, and it is the singleton on the simultaneous coverage event. □

C.2 Matrix confidence sequence and downstream projection

Proof of Theorem 4. For each row r, define the simultaneous row-coverage event

$$E_r := \{Q_r^\star \in \widetilde{\mathcal{C}}_{r,u}(\alpha_r) \text{ for every } u = 0, \ldots, N_r\}. \tag{47}$$

Theorem 3 gives $\mathbb{P}(E_r^c) \le \alpha_r$. Adaptive interleaving does not alter this statement: whenever row r is selected, the auditor reveals the next element of that row's pre-randomized order, so the observations from row r are a prefix of the same without-replacement sample path used in the row-local theorem. Predictable decisions to pause and resume the row merely select which local time $u_r(t)$ is displayed at global time t.

No independence between the events E_r is required. A union bound gives

$$\mathbb{P}\left(\bigcap_r E_r \right) \ge 1 - \sum_r \mathbb{P}(E_r^c) \ge 1 - \sum_r \alpha_r \ge 1 - \alpha. \tag{48}$$

On $\cap_r E_r$, every true row belongs to its displayed row set at every global time, and therefore $C^\star \in \mathcal{C}_t^C$ for every t. This proves (17). Because the event is simultaneous over all global times, it also implies $C^\star \in \mathcal{C}_\tau^C$ for any stopping time τ. □

Proof of Theorem 5. On the event in (17), $C^\star \in \mathcal{C}_t^C$ for every t. Consequently, for every t,

$$\min_{C \in \mathcal{C}_t^C} g(C) \le g(C^\star) \le \max_{C \in \mathcal{C}_t^C} g(C). \tag{49}$$

Thus the matrix-coverage event is contained in the simultaneous score-coverage event, and the latter has probability at least $1 - \alpha$. The statement at a stopping time follows from the same pathwise inclusion. □

Proof of Proposition 1. For any $C, C' \in \mathcal{C}_t^C$, Equation (21) and the triangle inequality imply

$$\begin{aligned} \|p(C) - p(C')\|_1 &= \left\| \sum_r w_r \{\theta_r(C) - \theta_r(C')\} \right\|_1 && (50)\\ &\le \sum_r w_r \|\theta_r(C) - \theta_r(C')\|_1 && (51)\\ &\le \sum_r w_r \delta_{r,u_r(t)}. && (52) \end{aligned}$$

Taking the supremum over C, C' proves (23). Moreover, the Lipschitz condition gives

$$\begin{aligned} \text{width}\{I_t(h)\} &= \sup_{C,C' \in \mathcal{C}_t^C} |h(p(C)) - h(p(C'))| && (53)\\ &\le L_h \sup_{C,C' \in \mathcal{C}_t^C} \|p(C) - p(C')\|_1. && (54) \end{aligned}$$

Combining the two displays proves (25). □

C.3 Biodiversity-pillar bounds

Proof of Proposition 2. Every species counted by $L_{1,t}$ has at least one expert-verified occurrence in the dataset, so it is present in the true corrected data. Therefore $L_{1,t} \le P_1(C^\star)$ deterministically.

For the first-hit calculation, a size-u uniform sample from a pool with n_s records and m_s successes contains no success with probability

$$\mathbb{P}(T_s > u) = \frac{\binom{n_s - m_s}{u}}{\binom{n_s}{u}} = \frac{\binom{n_s - u}{m_s}}{\binom{n_s}{m_s}}. \tag{55}$$

Taking the complement proves the first expression in (26). For the expectation, use the tail-sum formula and the hockey-stick identity:

$$\begin{aligned} \mathbb{E}[T_s] &= \sum_{u=0}^{n_s - m_s} \mathbb{P}(T_s > u) && (56)\\ &= \frac{1}{\binom{n_s}{m_s}} \sum_{u=0}^{n_s - m_s} \binom{n_s - u}{m_s} && (57)\\ &= \frac{\binom{n_s+1}{m_s+1}}{\binom{n_s}{m_s}} = \frac{n_s + 1}{m_s + 1}. && (58) \end{aligned}$$

□

Proof of Theorem 6. Fix a target group G, and write $x = p_G$, $y = p'_G$, $a = \|x\|_1 = m_G(p)$, and $b = \|y\|_1 = m_G(p')$. Because $a, b \ge \mu_G$,

$$\begin{aligned} \left\| \frac{x}{a} - \frac{y}{b} \right\|_1 &\le \left\| \frac{x - y}{a} \right\|_1 + \left\| y \left(\frac{1}{a} - \frac{1}{b} \right) \right\|_1 && (59)\\ &= \frac{\|x - y\|_1}{a} + \frac{|b - a|}{a} && (60)\\ &\le \frac{2}{\mu_G} \|x - y\|_1 \le \frac{2}{\mu_G} \|p - p'\|_1. && (61) \end{aligned}$$

For q on the fixed active support,

$$\frac{\partial D_1(q)}{\partial q_s} = -D_1(q)(1 + \log q_s). \tag{62}$$

Because $D_1(q) \le d_G$ and every active $q_s \ge \eta_G$,

$$\|\nabla D_1(q)\|_\infty \le d_G \max\{1, |1 + \log \eta_G|\}. \tag{63}$$

The line segment joining $q_G(p)$ and $q_G(p')$ remains in the simplex, has the same active support, and retains the coordinate lower bound η_G. The mean-value inequality therefore gives

$$|D_1(q_G(p)) - D_1(q_G(p'))| \le d_G \max\{1, |1 + \log \eta_G|\}\|q_G(p) - q_G(p')\|_1 \tag{64}$$

$$\le \frac{2d_G}{\mu_G} \max\{1, |1 + \log \eta_G|\}\|p - p'\|_1. \tag{65}$$

This proves (29). Summing the groupwise inequalities and applying the triangle inequality proves (30). □

Proof of Theorem 7. Let W be the symmetric matrix with off-diagonal entries ω_{ij} and zero diagonal. Then

$$A(q) = \frac{1}{2}q^\top W q, \qquad B(q) = \frac{1}{2}\{1 - \|q\|_2^2\}. \tag{66}$$

For normalized compositions p, q,

$$|A(p) - A(q)| = \frac{1}{2}|(p - q)^\top W(p + q)| \tag{67}$$

$$\le \frac{1}{2}\|p - q\|_1\|W(p + q)\|_\infty \tag{68}$$

$$\le \Omega\|p - q\|_1, \tag{69}$$

because $\|p + q\|_1 = 2$ and every entry of W is at most Ω. Similarly,

$$|B(p) - B(q)| = \frac{1}{2}|(p - q)^\top(p + q)| \tag{70}$$

$$\le \frac{1}{2}\|p - q\|_1\|p + q\|_\infty \le \|p - q\|_1. \tag{71}$$

Also $0 \le A(q) \le \Omega B(q)$. If $B(p), B(q) \ge \beta$, then

$$\left|\frac{A(p)}{B(p)} - \frac{A(q)}{B(q)}\right| \le \frac{|A(p) - A(q)|}{B(p)} + \frac{A(q)|B(p) - B(q)|}{B(p)B(q)} \tag{72}$$

$$\le \frac{\Omega}{\beta}\|p - q\|_1 + \frac{\Omega B(q)}{\beta B(q)}\|p - q\|_1 \tag{73}$$

$$= \frac{2\Omega}{\beta}\|p - q\|_1, \tag{74}$$

which proves (32). □

Proof of Corollary 1. Theorems 6 and 7 imply, for any two plausible compositions,

$$|M_{\mathrm{sm}}(p) - M_{\mathrm{sm}}(p')| \le |\lambda_2|\,|P_2(p) - P_2(p')| + |\lambda_3|\,|P_3^{\mathrm{sm}}(p) - P_3^{\mathrm{sm}}(p')| \tag{75}$$

$$\le (|\lambda_2|L_2 + |\lambda_3|L_3)\|p - p'\|_1. \tag{76}$$

Apply Proposition 1 with $L_h = |\lambda_2|L_2 + |\lambda_3|L_3$ to obtain (34). □

D Priority-Aware Active Auditing

The confidence sequence in Section B permits the auditor to choose submitted-label rows adaptively, provided that the next record within the selected row is sampled uniformly without replacement. Validity therefore does not determine where the next expert review should be spent. We now develop a priority-aware sampling (PAS) policy that uses the downstream score to direct audits toward rows whose remaining uncertainty matters most.

The inferential and acquisition roles are deliberately separated. The exact PPR matrix set and its projection supply the confidence sequence and the final stopping certificate. Estimated priorities and planning radii only select the next row. Consequently, a poor priority estimate can reduce efficiency but cannot invalidate coverage.

D.1 Sequencing the richness and abundance audits

Pillar 1 determines which species are present, whereas the sharper local Pillar 2 bound in Theorem 6 assumes that the active support is fixed. When operationally feasible, we therefore audit Pillar 1 to completion before applying the fixed-support Pillar 2 rule. Here, completion means that every eligible species has been certified as either present or absent across all rows in which it could occur; merely reaching a contractual richness floor is insufficient. Once support is known, a verified occurrence gives every active species a strictly positive abundance lower bound. If some species remain unresolved, we retain exact projection or a support-robust entropy continuity bound instead of invoking the fixed-support Lipschitz constant.

D.2 Directional influence and a normalized width envelope

Let g_j be a differentiable pillar or payment score with a predeclared full-width tolerance $\tau_j > 0$. Let

$$\mathcal{P}_t := \{p(C) : C \in \mathcal{C}_t^C\} \tag{77}$$

be the plausible corrected-composition region and let $S_{r,t}$ be the set of verified destinations that remain possible for submitted row r. Define the normalized directional influence

$$\Lambda_{jr,t} := \frac{w_r}{2\tau_j} \sup_{p \in \mathcal{P}_t} \left\{ \max_{s \in S_{r,t}} \partial_s g_j(p) - \min_{s \in S_{r,t}} \partial_s g_j(p) \right\}. \tag{78}$$

The row weight $w_r = N_r/N$ measures how much corrected mass row r can move. The gradient range measures how differently the score responds to the plausible destinations of that mass. Hence a large row need not be important if all of its plausible corrections have nearly identical downstream effects.

Theorem 8 (Directional width decomposition)**.** *Suppose $\mathcal{P}_t$ is contained in a convex region on which g_j is differentiable. Then*

$$\frac{\operatorname{width}\{I_t(g_j)\}}{\tau_j} \le B_{j,t} := \sum_r \Lambda_{jr,t} \delta_{r,u_r(t)}, \tag{79}$$

where $\delta_{r,u}$ is the row confidence-set diameter from (22).

The factor $1/2$ in (78) exploits the fact that two candidate probability rows have the same total mass: their difference is a zero-sum vector. Theorem 8 is therefore sharper than a global Lipschitz bound that allows a row perturbation to move in an arbitrary direction. Proofs for this section are collected in Appendix F.

For multiple protected scores, one may monitor

$$B_t^{\max} := \max_j B_{j,t}, \tag{80}$$

which places the pillars on comparable scales through τ_j. A weighted sum of the $B_{j,t}$ may be used when the frozen payment program supplies explicit pillar weights.

D.3 A finite-population planning objective

Exact PPR projection can be expensive to recompute after every possible next audit. Query design therefore uses a fast forecast of row contraction. Let K_r be the number of eligible verified destinations for row r, let $K := \max_r K_r$, $N_{\max} := \max_r N_r$, and define, for $1 \le u \le N_r/2$,

$$b_{r,u} := \sqrt{\frac{\rho_{r,u}}{2u} \log\left(\frac{2RKN_{\max}}{\alpha}\right)}, \qquad \rho_{r,u} := 1 - \frac{u-1}{N_r}, \tag{81}$$

$$\bar{\delta}_r(u) := 2K_r b_{r,u}. \tag{82}$$

The reported row diameter is capped at the simplex diameter two. We retain the untruncated expression $\bar{\delta}_r$ for allocation because it is decreasing and convex before half-census. The logarithmic factor follows from a finite union over rows, destinations, and row-local audit times; sharper confidence sequences may be substituted without changing the allocation argument.

Proposition 3 (Monotone convex planning radius)**.** *For $1 \le u \le N_r/2$, $\bar{\delta}_r(u)$ is decreasing and convex. Consequently, for every fixed coefficient $\lambda_r \ge 0$, the discrete marginal reductions*

$$\Delta_r(u) := \lambda_r\{\bar{\delta}_r(u) - \bar{\delta}_r(u+1)\} \tag{83}$$

are nonincreasing in u.

Fix a planning epoch during which the coefficients λ_r are held constant. For a single protected score, λ_r may equal a conservative $\Lambda_{r,t}$; for several pillars it may be obtained from the maximum or a weighted aggregation of their normalized influences. Given a common burn-in m_r, row caps $M_r \le N_r/2$, and total audit budget B, consider

$$\min_{u_1,\dots,u_R} \sum_r \lambda_r \bar{\delta}_r(u_r) \quad \text{subject to} \quad \sum_r u_r = B, \qquad m_r \le u_r \le M_r, \quad u_r \in \mathbb{Z}. \tag{84}$$

Theorem 9 (Greedy optimality for the frozen planning objective)**.** *Under Proposition 3, repeatedly selecting*

$$R_{t+1} \in \arg\max_{r: u_r < M_r} \lambda_r\{\bar{\delta}_r(u_r) - \bar{\delta}_r(u_r+1)\} \tag{85}$$

solves (84) globally for every feasible integer budget B.

The theorem concerns the fixed separable planning objective, not the random exact nonlinear projection width. It supplies an exact solution to a tractable surrogate. When $\bar{\delta}_r(u) \approx \kappa_r u^{-1/2}$, the continuous relaxation has a transparent allocation.

Corollary 2 (Two-thirds allocation)**.** *Put $d_r := \lambda_r\kappa_r > 0$. Ignoring row caps, the optimizer of*

$$\min_{u_r>0} \sum_r \frac{d_r}{\sqrt{u_r}} \quad \textit{subject to} \quad \sum_r u_r = B \tag{86}$$

is

$$u_r^* = B\frac{d_r^{2/3}}{\sum_k d_k^{2/3}}, \qquad V^*(B) = \frac{\left(\sum_k d_k^{2/3}\right)^{3/2}}{\sqrt{B}}. \tag{87}$$

If estimates satisfy $\gamma^{-1}d_r \le \widehat{d}_r \le \gamma d_r$ for every row, allocation according to $\widehat{d}_r^{2/3}$ inflates this leading width by at most $\gamma^{2/3}$.

Priority changes the leading constant, not the generic root-audit contraction rate. It is most useful when row influences are heterogeneous.

D.4 Support-aware mixed PAS

The conservative influence in (78) can be costly to optimize, and a plug-in influence that discards unobserved transitions can permanently under-prioritize rare but consequential corrections. Our primary implementable method is therefore *posterior support-presence PAS*. Under the positive Dirichlet–multinomial working prior from Section B, define the audit-live planning coefficient

$$\widehat{\Lambda}^{\mathrm{SP}}_{jr,t} := \mathbb{E}_{\pi_t}\left[\frac{w_r}{2\tau_j}\left\{\max_{s\in\operatorname{supp}(Q_r)} \partial_s g_j(p(Q)) - \min_{s\in\operatorname{supp}(Q_r)} \partial_s g_j(p(Q))\right\}\middle|\mathcal{F}_t\right]. \tag{88}$$

The expectation may be approximated by posterior draws. Because all prior parameters are positive, an unseen transition retains positive posterior probability and therefore cannot be removed merely because it was absent from the pilot. Equation (88) is a planning heuristic, not a confidence-width certificate.

At the beginning of the audit, we draw a randomized pilot

$$m_r := \min\{N_r, \max(5, \lceil\sqrt{N_r}\rceil)\} \tag{89}$$

from every row. At geometrically increasing epochs, we update the posterior, the point estimate of the corrected composition, and the priority coefficients. Within an epoch, coefficients are frozen and the priority branch applies the greedy rule (85).

To protect against model misspecification and rare real-data transitions, the priority branch can be mixed with record-level simple random sampling. Let $\varepsilon \in [0, 1]$ be the SRS floor. Conditional on the current history, the next row is drawn according to

$$\mathbb{P}(R_{t+1} = r \mid \mathcal{F}_t) = \varepsilon \frac{N_r - u_r(t)}{N - t} + (1 - \varepsilon)\mathbb{1}\{r = r_t^{\mathrm{PAS}}\}, \tag{90}$$

where r_t^{PAS} maximizes the current priority marginal gain. After choosing a row, the algorithm always reveals the next record in that row's pre-randomized order. We evaluate pure support-presence PAS and mixtures with $\varepsilon = 0.75$ and 0.90.

Proposition 4 (Validity of support-aware mixed PAS)**.** *For any predictable audit-live priority estimates, epoch schedule, and SRS floor ε, support-aware mixed PAS preserves the simultaneous matrix, pillar, and payment coverage guarantees of Theorems 4 and 5.*

The reason is structural: the algorithm chooses only the row, before observing the next verified label, and never alters the randomized order within that row.

Stopping. Let $[L_{j,t}, U_{j,t}] = I_t(g_j)$. The auditor stops when a predeclared decision is certified, for example when $L_{j,t} \geq \gamma_j$ or $U_{j,t} < \gamma_j$, or when every protected score satisfies $U_{j,t} - L_{j,t} \leq \tau_j$. If no statistical criterion is reached before a maximum review budget, the outcome is reported as inconclusive. Neither the plug-in priority, the planning surrogate, nor point-estimation error is used as a live stopping rule.

E Experiments and Results

We evaluate three distinct questions: whether the categorical PPR construction is anytime-valid; whether priority-aware allocation can reduce the number of audits needed for a fixed downstream target when influences are heterogeneous; and whether implementable PAS variants learn useful priorities on a realistic, high-dimensional correction matrix. We report certified-width and point-estimation experiments separately.

E.1 Natural State data and replay design

The Natural State dataset contains 33,403 auditable records from a biodiversity-credit project in central Kenya. It has 46 submitted categories, 32 categories after expert correction, and 50 categories in the union of the two vocabularies. The completed correction matrix contains 220 nonzero cells, of which 192 are off diagonal. In total, 6,002 records are corrected, an off-diagonal rate of 17.97%. Row sizes range from 2 to 12,075 records, and 20 of the 46 rows contain fewer than 20 records. The five largest directional corrections account for 64.1% of all corrections. This combination of extreme row imbalance, concentrated common errors, and rare transitions motivates both priority sampling and continued exploration.

Because complete adjudications are available for evaluation, we treat the matrix as a fixed finite population and replay an audit by independently randomizing the order within each submitted-label row. All policies on a paired path use the same underlying row orders. We run 100 paired replay paths and report means together with 10th–90th percentile bands through 5,000 expert audits.

We compare record-level SRS with five implementable policies: current plug-in PAS, posterior support-presence PAS, submitted-catalog plug-in PAS, and PAS with 75% or 90% SRS floors. Truth-informed allocation is retained only as an offline diagnostic and is not used in any practitioner-facing result.

E.2 Evaluation metrics

We use four metrics.

1. **Pillar 2 point-estimation MAE.** At each budget, we compare the plug-in Hill–Shannon diversity estimate with the full-data corrected value and average the absolute error across replay paths.

Table 2: Natural State replay results after 5,000 audits, averaged across 100 paired paths. Lower is better. The common surrogate and allocation distance are offline, non-certified diagnostics.

Policy	Common surrogate	Pillar 2 MAE	Allocation TV distance
Simple random sampling	28.60	0.0426	0.516
Current plug-in PAS	29.41	**0.0203**	0.215
Posterior support-presence PAS	**18.77**	0.0279	**0.140**
Submitted-catalog plug-in PAS	22.12	0.0328	0.328
PAS with 75% SRS floor	23.15	0.0347	0.403
PAS with 90% SRS floor	25.05	0.0353	0.458

2. **Common truth-weighted planning surrogate.** To compare allocations on a common scale, we evaluate every realized allocation using the same full-data directional coefficients in the Theorem 9 surrogate. These coefficients are used only after the replay for evaluation; they are not supplied to an implementable algorithm. This metric is neither a confidence sequence nor a proven upper bound on the exact projected PPR width.
3. **Allocation recovery.** We report total-variation distance between each terminal row-allocation vector and the allocation generated by a truth-informed reference policy. This is also an offline mechanism diagnostic.
4. **Certified PPR width.** On small categorical populations where complete enumeration is possible, we compute the exact projected PPR interval and the directional envelope at every audit time.

The first metric evaluates point estimation; the second and third diagnose the priority mechanism; only the fourth is a confidence-sequence width.

E.3 Natural State replay results

Table 2 reports the implementable policies at 5,000 audits. Every PAS method improves Pillar 2 point-estimation MAE relative to SRS. Current plug-in PAS attains the smallest terminal MAE, 0.0203 versus 0.0426 for SRS, but slightly worsens the common planning surrogate. Posterior support-presence PAS gives the strongest balanced result: it reduces the common surrogate by 34.4%, reduces MAE by 34.5%, and reduces allocation distance to the truth-informed reference by 72.9%. The 75% and 90% SRS mixtures retain smaller but consistent improvements, demonstrating that substantial random exploration can coexist with task-directed gains.

Using the SRS error after 5,000 audits as a matched-accuracy target gives a retrospective tolerance of 0.0426 effective-species units. Linear interpolation of the mean replay curves shows that posterior support-presence PAS reaches this MAE after approximately 2,520 audits, compared with 5,000 for SRS. It therefore audits 7.5% rather than 15.0% of the 33,403-record population and saves approximately 2,480 expert audits, or 49.6%. This is a matched-point-estimation comparison, not a live stopping rule: the true MAE is unobservable during an unfinished audit.

The mechanism diagnostic supports the intended explanation. Across the three direct priority-estimation policies, contribution-weighted priority error is strongly associated with common-surrogate regret (pooled Spearman $\rho = 0.92$). The relationship is descriptive rather than causal and uses full-data quantities unavailable to the auditor.

E.4 Anytime validity and synthetic acquisition efficiency

We first stress-test simultaneous coverage on 12 fixed finite populations, crossing $L \in \{3, 6, 12\}$, balanced versus Zipf-like row sizes, and uniform versus heterogeneous correction patterns. Each condition uses 3,000 complete randomized audit paths, for 36,000 paths per valid construction. At nominal 95% simultaneous coverage, worst full-path noncoverage is 3.17% for joint product inversion and 2.47% for rowwise Bonferroni inversion. Repeatedly inspecting ordinary fixed-time Wald intervals fails on 71.1% of paths. The PPR procedures therefore control optional stopping in regimes where repeated fixed-time intervals do not.

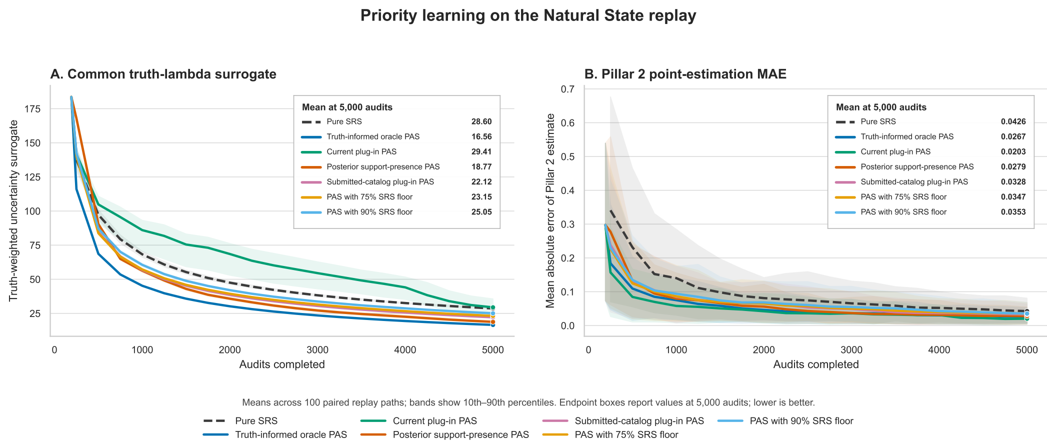

Figure 2: Natural State replay. Panel A evaluates each realized allocation with common full-data coefficients and is non-certified. Panel B reports Pillar 2 point-estimation MAE. Curves are means over 100 paired paths and bands show 10th–90th percentiles.

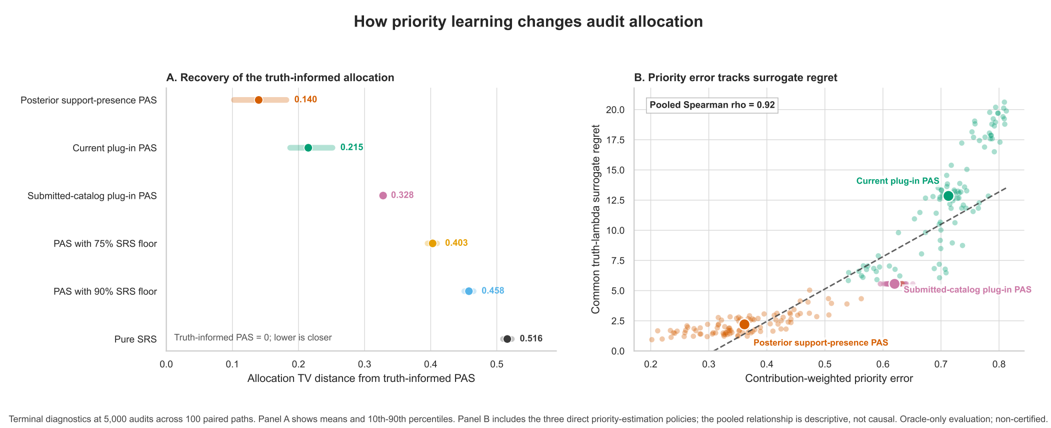

Figure 3: Offline mechanism diagnostics at 5,000 audits. Lower allocation distance and priority error indicate closer recovery of the truth-informed reference. These quantities diagnose acquisition but are not available as live audit certificates.

In the heterogeneous-influence acquisition experiment, the fixed population contains 5,700 records in eight rows and every policy receives a 20-per-row pilot. The stopping target is 35% of the common post-pilot exact designed smooth-score width. Learn–freeze influence reaches the target after 533 audits, compared with a median 996 for random-row selection, 657 for largest marginal width decrease, and 1,451 for largest current width. Relative to random rows, priority learning reduces audits by 46.5%. At stopping, the local directional envelope is 3.7% wider than the exact designed interval.

The gain is not universal. Under homogeneous row influence, priority learning improves on random rows by only 2%. In an exactly enumerated $N = 54$ categorical-PPR negative control, random selection is slightly better at several early budgets. Priority-aware auditing improves the leading constant when downstream influence is heterogeneous and the envelope is sharp; it does not change the generic root-audit rate or dominate SRS in every regime.

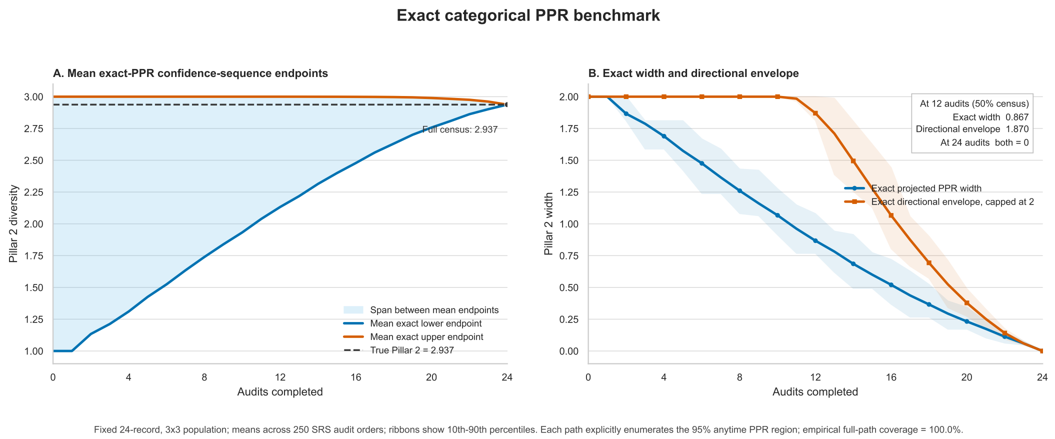

Figure 4: Exact categorical-PPR benchmark. The exact projected Pillar 2 interval and the directional envelope contract to zero at census; the envelope is conservative at intermediate budgets.

E.5 Exact categorical-PPR benchmark

To connect the theory to an explicitly calculable confidence sequence, we use a fixed 24-record, 3×3 confusion matrix and enumerate the 95% PPR region at every audit time over 250 SRS orders. The true Pillar 2 value is 2.937. At 12 audits, corresponding to 50% census, the mean exact projected width is 0.867 and the mean directional envelope is 1.870. Both equal zero at the full 24-record census, and empirical full-path coverage is 100% across the 250 orders. The example verifies contraction and eventual collapse while also showing that the directional envelope can be conservative early in the audit.

E.6 Scope of the evidence

The Natural State replay demonstrates implementable acquisition and point-estimation gains, but exact categorical PPR inversion is combinatorial at its full scale. The real-data surrogate is not a confidence sequence, and the computable valid upper-bound experiment shows little separation between PAS and SRS. We therefore do not claim a real-data reduction in valid projected width or audits required for certification. The certified evidence consists of the coverage stress test, the exact small-population benchmark, and the synthetic designed-width experiment. Establishing end-to-end certified savings on a prospective field audit remains future work.

F Proofs for Priority-Aware Auditing

This appendix proves the deterministic sensitivity and allocation results used in Section D. Throughout, all quantities at the audit time under consideration are treated as fixed unless explicitly stated otherwise.

F.1 A zero-sum range inequality

Lemma 2 (Range bound for probability contrasts)**.** *Let $a, d \in \mathbb{R}^K$, suppose $\sum_{k=1}^{K} d_k = 0$, and let $S \subseteq \{1, \ldots, K\}$ contain the support of d. Then*

$$|a^\top d| \leq \frac{1}{2} \left\{ \max_{k \in S} a_k - \min_{k \in S} a_k \right\} \|d\|_1. \tag{91}$$

Proof. Write $d = d^+ - d^-$, where $d_k^+ = \max(d_k, 0)$ and $d_k^- = \max(-d_k, 0)$. The zero-sum condition implies

$$\sum_k d_k^+ = \sum_k d_k^- =: m = \frac{1}{2}\|d\|_1.$$

If $m = 0$, the claim is immediate. Otherwise, $q^+ = d^+/m$ and $q^- = d^-/m$ are probability vectors supported on S, so

$$a^\top d = m\{a^\top q^+ - a^\top q^-\}.$$

Both expectations lie between $\min_{k \in S} a_k$ and $\max_{k \in S} a_k$. Their absolute difference is therefore at most the range of a on S, which proves (91). $\square$

F.2 Proof of the directional width bound

Proof of Theorem 8. Take arbitrary $C, C' \in \mathcal{C}_t^C$, and write their rows as $\theta_1, \ldots, \theta_R$ and $\theta'_1, \ldots, \theta'_R$. Construct intermediate matrices $C^{(0)}, \ldots, C^{(R)}$ by setting $C^{(0)} = C$, $C^{(R)} = C'$, and replacing row r at step r. If $p^{(r)} = p(C^{(r)})$, then

$$p^{(r)} - p^{(r-1)} = w_r(\theta'_r - \theta_r). \tag{92}$$

The vector $d_r := \theta'_r - \theta_r$ is supported on $S_{r,t}$ and satisfies $\mathbf{1}^\top d_r = 0$, because both rows are probability vectors.

By the fundamental theorem of calculus along the line segment between $p^{(r-1)}$ and $p^{(r)}$,

$$g_j(p^{(r)}) - g_j(p^{(r-1)}) = w_r \int_0^1 \nabla g_j\Big(p^{(r-1)} + z\{p^{(r)} - p^{(r-1)}\}\Big)^\top d_r \, \mathrm{d}z.$$

The entire segment lies in the assumed convex differentiability region. Applying Lemma 2 pointwise in z, and then the definition of $\Lambda_{jr,t}$, gives

$$\begin{aligned}\Big|g_j(p^{(r)}) - g_j(p^{(r-1)})\Big| &\le \frac{w_r}{2} \sup_{p \in \mathcal{P}_t} \left\{\max_{s \in S_{r,t}} \partial_s g_j(p) - \min_{s \in S_{r,t}} \partial_s g_j(p)\right\} \|\theta'_r - \theta_r\|_1 \\ &= \tau_j \Lambda_{jr,t} \|\theta'_r - \theta_r\|_1.\end{aligned}$$

The row confidence-set diameter implies $\|\theta'_r - \theta_r\|_1 \le \delta_{r,u_r(t)}$. Telescoping over the intermediate matrices and applying the triangle inequality therefore yields

$$|g_j(p(C')) - g_j(p(C))| \le \tau_j \sum_r \Lambda_{jr,t} \delta_{r,u_r(t)}.$$

Taking the supremum over $C, C' \in \mathcal{C}_t^C$ gives exactly the width of the projected interval $I_t(g_j)$, and proves (79). $\square$

F.3 Shape of the finite-population planning radius

Proof of Proposition 3. All multiplicative factors in $\bar{\delta}_r(u)$ are positive constants with respect to u, so it is enough to study

$$f(u) := \sqrt{\frac{a}{u} - b}, \qquad a := \frac{N_r + 1}{N_r}, \qquad b := \frac{1}{N_r}.$$

For $0 < u < N_r + 1$, direct differentiation gives

$$f'(u) = -\frac{a}{2u^2\sqrt{a/u - b}} < 0, \tag{93}$$

$$f''(u) = \frac{a(3a - 4bu)}{4u^4(a/u - b)^{3/2}}. \tag{94}$$

If $u \le N_r/2$, then $3a - 4bu \ge 3(N_r + 1)/N_r - 2 > 0$, so $f''(u) > 0$. Hence $\bar{\delta}_r$ is decreasing and convex on the stated interval.

For any convex function, its forward increments $\bar{\delta}_r(u+1) - \bar{\delta}_r(u)$ are nondecreasing in u. Multiplying their negatives by $\lambda_r \ge 0$ shows that $\Delta_r(u) = \lambda_r\{\bar{\delta}_r(u) - \bar{\delta}_r(u+1)\}$ is nonincreasing. $\square$

F.4 Optimality of greedy allocation

Proof of Theorem 9. Start with the mandatory burn-in allocation $(m_1, \ldots, m_R)$. For each row r, form the ordered list of available marginal reductions

$$\Delta_r(m_r), \Delta_r(m_r + 1), \ldots, \Delta_r(M_r - 1).$$

Proposition 3 shows that every list is nonincreasing. Assigning $k_r = u_r - m_r$ additional audits to row r selects exactly the first k_r entries of its list, and the total decrease from the burn-in objective is their sum. Thus any feasible allocation of $B - \sum_r m_r$ additional audits corresponds to selecting that many marginal reductions subject only to the prefix constraint within each row.

At every step, rule (85) selects the largest currently available head among the row lists. Because the lists are nonincreasing, this is also a largest unselected feasible marginal reduction. Suppose a feasible solution differed from the greedy solution at the first greedy selection it omitted. It must instead contain a weakly smaller marginal reduction. Swap that selected reduction for the omitted greedy reduction. The prefix constraint remains satisfied: every predecessor of the greedy item was chosen at an earlier greedy step, while removing the last selected item from the other row preserves its prefix. The swap cannot reduce the total objective decrease. Repeating this exchange transforms an optimal solution into the greedy solution. Therefore the greedy allocation is globally optimal. □

F.5 The two-thirds rule and robustness to priority error

Proof of Corollary 2. The objective is strictly convex on the positive orthant. With multiplier ν for $\sum_r u_r = B$, the first-order conditions are

$$-\frac{d_r}{2u_r^{3/2}} + \nu = 0, \qquad r = 1, \ldots, R.$$

Thus $u_r = (d_r/(2\nu))^{2/3}$. Enforcing the budget constraint yields

$$u_r^* = B\frac{d_r^{2/3}}{\sum_k d_k^{2/3}}.$$

Substitution into the objective gives

$$V^*(B) = B^{-1/2}\left(\sum_k d_k^{2/3}\right)^{3/2}.$$

Now let $\widehat{u}_r = B\widehat{d}_r^{2/3}/\sum_k \widehat{d}_k^{2/3}$. The true objective attained by this estimated-priority allocation is

$$V(\widehat{u}) = B^{-1/2}\left(\sum_k \widehat{d}_k^{2/3}\right)^{1/2} \sum_r d_r \widehat{d}_r^{-1/3}. \tag{95}$$

The multiplicative error assumption implies

$$\left(\sum_k \widehat{d}_k^{2/3}\right)^{1/2} \le \gamma^{1/3}\left(\sum_k d_k^{2/3}\right)^{1/2},$$
$$\sum_r d_r \widehat{d}_r^{-1/3} \le \gamma^{1/3} \sum_r d_r^{2/3}.$$

Combining these inequalities in (95) gives $V(\widehat{u}) \le \gamma^{2/3} V^*(B)$, as claimed. □

F.6 Validity under adaptive support-aware acquisition

Proof of Proposition 4. At time t, the posterior, the priority coefficients, the epoch state, and the remaining row sizes are all $\mathcal{F}_t$-measurable. Hence the row distribution in (90) is predictable. The SRS-branch coin and any randomized tie-breaking are drawn independently of the unseen verified labels conditional on $\mathcal{F}_t$. Once a row is selected, the next record is the next element of that row's independently pre-randomized permutation. Consequently, conditional on the selected row and the history, the revealed record is uniform among that row's remaining records.

These are exactly the conditions required by the rowwise without-replacement martingale construction in Lemma 1. Therefore every row confidence sequence remains valid at all of its adaptively chosen local audit times. The simultaneous matrix guarantee then follows from Theorem 4. Finally, the pillar and payment intervals are deterministic projections of the matrix set, so their simultaneous coverage follows from Theorem 5. No step of this argument assumes that the posterior working model or its priority estimates are correct. □

G Additional finite-population results and operational guidance

G.1 Binary mislabel-rate special case

The categorical construction contains a simple binary confidence sequence as a special case. Let fixed indicators $Z_i \in \{0,1\}$ mark whether each of N records is mislabeled, let $F = \sum_i Z_i$, and expose a uniformly randomized order $X_1, \ldots, X_N$. With $S_t = \sum_{j \le t} X_j$,

$$\mathbb{P}(X_t = 1 \mid \mathcal{F}_{t-1}) = \frac{F - S_{t-1}}{N - t + 1}. \tag{96}$$

For a beta–binomial working prior $F \sim \mathrm{BetaBin}(N, a, b)$, the PPR at candidate F_0 is

$$R_t(F_0) = \frac{\binom{N}{F_0}}{\binom{N-t}{F_0 - S_t}} \frac{B(a + S_t, b + t - S_t)}{B(a, b)}. \tag{97}$$

Thus $\{F_0 : R_t(F_0) < 1/\alpha\}$ is an anytime-valid finite-population confidence sequence. Sampling without replacement contracts faster than sampling with replacement because of the finite-population correction; for an approximate common variance target, $t_{\mathrm{wor}} = t_{\mathrm{wr}}/(1 + t_{\mathrm{wr}}/N)$.

G.2 Heterogeneous audit costs and Neyman allocation

For the continuous width objective $\sum_r d_r/\sqrt{u_r}$ under $\sum_r c_r u_r = B$, the Lagrange conditions give

$$u_r^* = B \frac{d_r^{2/3} c_r^{-2/3}}{\sum_j d_j^{2/3} c_j^{1/3}}. \tag{98}$$

This differs from Neyman allocation for minimizing first-order point-estimator variance, which (ignoring finite-population corrections) allocates $u_r \propto w_r \sigma_{r,g}/\sqrt{c_r}$. The two-thirds rule minimizes a sum of confidence-width contributions; Neyman allocation minimizes a sum of variances.

G.3 Richness query guarantees

If a candidate pool of size n_s contains $m_s \ge 1$ true occurrences, the probability of no hit after u uniform audits is

$$\frac{\binom{n_s - m_s}{u}}{\binom{n_s}{u}} \le \left(1 - \frac{m_s}{n_s}\right)^u \le e^{-u m_s / n_s}. \tag{99}$$

For a richness-only objective, continuing to sample the presence-search pool of a species after its first verified hit is pathwise weakly dominated by redirecting those audits to unresolved species. If remaining hit probabilities were known, the largest one maximizes the one-step chance of adding a richness unit; unknown hazards motivate a model score plus exploration bonus. This is a one-step statement, not global finite-horizon optimality.

G.4 Operational stopping and tolerances

Tolerance τ_j should be the largest uncertainty that cannot change a credit, payment, or approval decision. The auditor predeclares α, scoring software, decision thresholds, pillar or payment tolerances, and a maximum budget; runs a randomized pilot; lets PAS select rows while preserving within-row randomization; and recomputes a valid projected interval at prespecified checkpoints. Stop for certification ($L_{j,t} \geq \gamma_j$), rejection ($U_{j,t} < \gamma_j$), precision ($U_{j,t} - L_{j,t} \leq \tau_j$ for all protected scores), or census. Hitting the maximum budget without a statistical condition is reported as inconclusive. A practical tolerance grid is 10%, 5%, 2%, and 1% of each pillar's operational range, with 5% primary; the final operational target should be an immaterial payment error rather than a convenient simulation width.